# PathMNIST Federated Learning: FedAvg vs FedProx Under Non-IID Label Noise

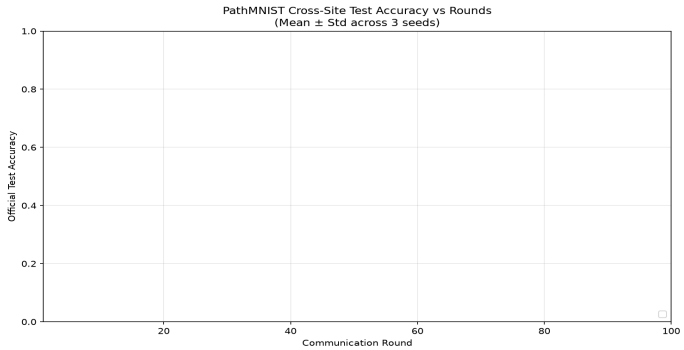
Authoritative Benchmark & Evaluation Report | Seeds: 42, 123, 2026 | Dirichlet $\alpha$=0.3 | 40% Symmetric Noise on Client 0

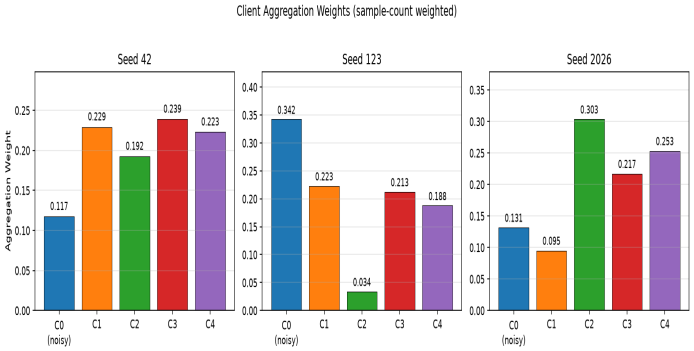
**Experimental Design:** 5-client Dirichlet ($\alpha$=0.3) partitioning over official PathMNIST training data (89,996 images). Client 0 is subjected to 40% symmetric label noise on its local training set. 100 communication rounds, 5 local epochs (E=5), batch size 64, SGD optimizer (lr=0.01, momentum=0.9, fresh optimizer state per round). FedProx proximal term evaluated against detached start-of-round snapshot. Evaluation strictly isolates official in-distribution validation (10,004) from clinical cross-site test (7,180).

## Table 1: Cross-Site Official Test & Validation Performance (Mean ± Std Across 3 Seeds)

| Experimental Group | Test Acc (R100) | Val Acc (R100) | Test Acc (91-100) | Val Acc (91-100) | C0 Holdout (R100) |
|---|---|---|---|---|---|
| **FEDAVG clean** | N/A | N/A | N/A | N/A | N/A |
| **FEDAVG noisy** | N/A | N/A | N/A | N/A | N/A |
| **FEDPROX clean** | N/A | N/A | N/A | N/A | N/A |
| **FEDPROX noisy** | N/A | N/A | N/A | N/A | N/A |

## Key Visualizations: Convergence and Diagnostic Metrics




**Update-Metric Observations & Robustness:**

• *Gradient Norms & Cosine Alignment:* Client 0 (40% noisy) exhibits altered update dynamics. In non-IID partitioning ($\alpha$=0.3), low cosine alignment naturally occurs due to class imbalance; however, Client 0's leave-one-out cosine shows systematic divergence from clean clients.

• *Aggregation Weight Sensitivity:* Client 0's influence is directly governed by its sample-count aggregation weight (p0). Across seeds 42, 123, and 2026, p0 values are 0.117, 0.342, and 0.132 respectively. Noise impact is notably amplified under seed 123.

• *Proximal Term Behavior:* FedProx constrains drift from the global snapshot. When SGD momentum (0.9) is active, the effective proximal penalty interacts with velocity accumulation, stabilising client parameter trajectories without degenerating local learning.

**Limitations & Scientific Disclosures:** (1) PathMNIST official test represents clinical cross-site evaluation; cross-site accuracy is lower than in-distribution validation. (2) $\mu$ was selected using in-distribution validation over rounds 91-100 on seed 42. (3) Three seeds provide preliminary variance estimation without formal statistical significance claims. (4) Offline JSON logging utilized as primary reproducible audit trail due to environment connectivity constraints.